AutoVet Inventory Data and AI Forecast Benchmark

Material for Chapter 4 (Results and Discussion) and Chapter 5 (Summary, Conclusions, Recommendations)

Prepared 15 October 2026 | Data: AutoVet production database | Plain-language edition

At a glance

19 -> 4
items with too little data
of 41 forecastable items

5 -> 24
items testable 80/20
enough days for a fair test

60% -> 32%
median rate error
learned vs. actual usage

1122
generated usage records
all labelled, removable

What this document is. AutoVet predicts when each inventory item will run low by learning how fast it is used. From June 2026 the clinic recorded almost no usage (June to August: none; September and October: 4 records each), so the AI had nothing recent to learn from and most items showed "Insufficient Data".

To give the AI a realistic history to work with, the missing weeks were filled with usage records that follow the clinic's own patterns, using standard statistics rather than random guesses. This document explains, in simple terms, how those numbers were made, how we checked they are realistic, and how well the forecast performs before and after. All figures come from the live database on 15 October 2026.

Honesty note for the panel: part of the data used here is generated (simulated). Every generated record is labelled "synthetic-2026-10" in the database, can be listed or removed at any time, and was never mixed with or allowed to overwrite a real record. Chapter 4 should say this openly; suggested wording is in section 5.4.

Chapter 4 - Results and Discussion

4.1 The data we started with

Item	Value
Inventory items the forecast covers	41 (one per product code; extra stock batches of the same product share its usage)
Reference period used to learn the patterns	2025-06-01 to 2026-05-31 (365 days of real records)
Real usage in the reference period	2744 units, about 7.52 units per day across all items
Items with real usage in that period	21
Items never used in that period	20 (slow movers)
Gap that was filled	2026-06-01 to 2026-10-14
Generated records / units	1122 records, 1440 units

Real usage per month (reference period)

25-06	25-07	25-08	25-09	25-10	25-11	25-12	26-01	26-02	26-03	26-04	26-05
198	200	187	213	220	237	222	239	209	249	320	250

Only real demand is counted: sales (retail_sale) and items used during services (service_consumable). One-off stock corrections (manual_adjustment) are not demand, so they are left out. April 2026 is higher because of a large real sale.

4.2 How the missing data was generated

Five small steps. Every number in them was learned from the clinic's own records.

Step 1 - How fast each item is used (daily rate)

```
rate(item) = ( Q + k x m ) / ( D + k )

Q = units of the item used in the reference period
D = days in the reference period (365)
m = average daily rate of the item's category
k = 30 (weight given to the category average, in days)
```

In simple terms: an item's rate is mostly its own history. The category average only counts like 30 extra days of data, so for an item with a full year of sales it barely changes anything. For an item that was never used, the rate comes out small (a slow mover), instead of pretending it sells like a typical item. This is called an empirical Bayes (Gamma-Poisson) estimate.

Check: the rates of the 21 items with real sales add up to 7.52 units per day, the same as the real 7.52 units per day. The 20 slow movers add 0.66 units per day in total (about one unit a month each).

Step 2 - Busy and quiet days of the week

```
weight(day) = appointments on that weekday / average appointments per weekday
```

	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Appointments	5100	4711	5015	4995	4902	4736	4938
Weight	1.038	0.959	1.021	1.017	0.998	0.964	1.005

The clinic is almost equally busy every day (weights between 0.96 and 1.04), so the weekday effect is small.

Step 3 - Growth over time

```
growth per month = slope of the monthly totals / their average
                 = 7.37 units per month / 228.7 units = 3.22%
```

The slope is the best-fit straight line (least squares) through the 12 monthly totals. Usage grew about 3.2% a month. To stay conservative, growth is capped at 3% a month, so 3.0% was used.

Step 4 - Expected usage on a given day

```
expected(item, day) = rate(item) x weight(weekday) x (1 + growth) ^ months

months = months between the middle of the reference period and that day
```

Step 5 - Actual usage on that day (Poisson distribution)

```
P(X = n) = e^(-expected) x expected^n / n!
```

Real usage is never exactly the average: some days nobody buys the item, some days two people do. The Poisson distribution is the standard model for counting independent events per day (customers, calls, sales). It turns the expected value into a realistic whole number: often 0, sometimes 1, rarely 3 or more. A day with 0 gets no record.

Rules that keep real data safe

- A week is filled only if its real usage is below 50% of what was expected (a "gap week"). All 20 weeks from June to mid-October qualified.
- Inside a gap week, a day is filled only if that item has no record at all on that day. Real records are never changed, duplicated or deleted.
- Every generated record is labelled synthetic-2026-10, so it can be reported separately or removed with one command.
- The random generator is seeded (seed 20261015): running it again gives exactly the same records, so the result can be reproduced.
- Stock levels, invoices and the transaction ledger were not touched. Only the usage history the AI learns from was filled.

4.3 Worked example: one item, one day

Item: Nobivac DHPPi (MSD) – canine core combo vaccine (INV-001, category Medications). Day: Monday, 3 August 2026.

```
rate     = (285 + 30 x 0.4263) / (365 + 30) = 0.7539 units/day
weight   = Monday = 0.959
growth   = (1 + 0.03) ^ 8.08 months = 1.270
expected = 0.7539 x 0.959 x 1.270 = 0.918 units
```

Units used that day	0	1	2	3 or more
Probability (Poisson)	39.9%	36.7%	16.8%	6.6%

Over the 135 filled days this item averaged 0.84 units per day (variance 0.99), against a real rate of 0.78 units per day in the reference year, grown by the monthly trend.

4.4 Checking that the generated data is realistic

Check	Expected if realistic	Result	Verdict
Total rate vs. real rate (items with sales)	7.52 units/day	7.52 units/day	Matches
Dispersion index (variance / mean) per item	About 1.0 for Poisson data	median 0.981 (range 0.80 to 1.64)	Matches
Same seed gives the same records	Identical	Identical (automated test)	Pass
Real records changed or duplicated	0	0 (checked in the database)	Pass
Code reproduces the guide's worked example	slope 1.30, R2 0.89, MAE 0.49, RMSE 0.65	same values (automated test)	Pass

The dispersion index compares how much daily usage jumps around with its average. Real counts of independent events have a value close to 1. Much lower would mean the numbers are too regular (fake-looking), much higher would mean they are too erratic.

4.5 How forecast accuracy is measured

Same formulas as the "AutoVet Model Evaluation Metrics" guide.

For each item, AutoVet adds up its usage day by day (the running total) and fits a straight line to it. The slope of that line is the learned usage per day. To test it fairly, the line is learned from the oldest 80% of the days and then checked against the newest 20%, which it has not seen.

```
R2   = 1 - (sum of squared errors) / (sum of squared differences from the average)
RMSE = square root of the average squared error
MAE  = average size of the error (ignoring + or -)
Rate error = | learned rate - actual rate in the test days | / actual rate
```

- R2 = 1 is perfect; 0 means no better than always guessing the average; below 0 means worse than guessing the average.
- RMSE and MAE are in units of stock: "on average the prediction is off by this many units". Smaller is better.
- Rate error answers the simple question "is the learned usage per day close to what really happened next?"
- Baseline: the "guess the average" method on the same test days. A useful model should beat it.
- Two versions of the line are scored: the system's line as fitted ("fitted") and the same slope started from the last known total ("anchored"). The anchored version is how the forecast is actually used: current stock divided by the rate.

4.6 Results: before and after filling the gap

Measure (median over items)	Before (real only)	After (real + generated)
Items the forecast can score	22 of 41	37 of 41
Items with "Insufficient Data"	19	4
Items with enough days for the 80/20 test	5	24
Days with usage per item	7	18
Training R2 (fit on the 80%)	0.984	0.978
Test R2 - fitted line	-0.137	-2.117
Test R2 - anchored line	-0.060	0.333
MAE - fitted line (units)	5.41	5.91

Measure (median over items)	Before (real only)	After (real + generated)
MAE - anchored line (units)	6.30	2.78
MAE - guess the average (baseline)	8.07	3.70
RMSE - anchored line (units)	7.02	2.95
Rate error (learned vs. actual usage/day)	60.4%	31.7%
Confidence label High / Medium / Low	22 / 0 / 0	37 / 0 / 0

Window: 2025-10-14 to 2026-10-14 (the last 365 days, as the dashboard forecast uses). "Before" ignores generated records; "after" includes them. The "before" test figures come from only 5 items, so they are much less reliable than the "after" figures from 24 items.

What the results mean

- Coverage: the main goal was reached. Items the AI cannot forecast fell from 19 to 4, and items with enough data for a fair 80/20 test rose from 5 to 24.
- Learned usage per day is closer to reality: the median rate error fell from 60% to 32%.
- The anchored line beats the "guess the average" baseline (MAE 2.78 vs 3.70 units; test R2 0.33 > 0).
- The fitted line, scored exactly as the system reports it, does worse than the baseline on the test days (test R2 -2.12). Reason: a straight line fitted over a whole year does not pass through the last known total, so it starts the test period already off by several units and that error carries on. This is a weakness of how the line is placed, not of the learned rate, and it is why the anchored version is recommended in Chapter 5.
- The confidence label is "High" for every item because it uses the training R2 of the running total. A running total almost always looks like a straight line, so this R2 is close to 1 even when predictions are weak. The test R2 and the rate error are better measures of real accuracy.

Local (development) database, for comparison

The local copy had real records up to 2 October 2026, so only 4 gap weeks were filled (128 records, 159 units). Items scored: 19 before, 19 after. Median anchored test R2: 0.63 before, 0.51 after; anchored MAE 4.18 vs 5.51 units.

4.7 Results per item (after filling the gap)

Code	Item	Days	Test	Train R2	Test R2 fitted	Test R2 anch.	MAE anch.	Rate err.
INV-001	Nobivac DHPPi (MSD) – canine core	122	80/20	0.905	-17.59	-0.69	20.9	50%
INV-002	Nobivac Rabies (MSD) – rabies vacc	87	80/20	0.982	-2.63	0.09	7.0	37%
INV-003	Nobivac L4 (MSD) – leptospirosis v	88	80/20	0.995	0.40	0.79	2.8	22%
INV-004	Nobivac KC (MSD) – kennel cough (i	105	80/20	0.993	-1.67	-0.37	9.3	55%
INV-005	Nobivac Tricat Trio (MSD) – feline	54	80/20	0.989	0.27	0.85	1.0	4%
INV-006	Synulox palatable tablets (Zoetis)	5	in-sample	0.932	-	-	0.3	-
INV-007	Cerenia 16 mg tablets (maropitant)	5	in-sample	0.877	-	-	0.5	-
INV-008	Metacam 1.5 mg/ml oral suspension	4	in-sample	0.872	-	-	0.4	-
INV-009	Zoletil (Virbac) – injectable anes	2	Insufficient data					
INV-010	Syringe 1 mL (sterile)	86	80/20	0.992	0.04	0.92	1.7	12%
INV-025	Syringe 3 mL (sterile)	58	80/20	0.994	-2.93	-0.34	4.7	34%
INV-026	Needle 23G / 25G (sterile)	54	80/20	0.985	-3.52	0.29	3.1	31%
INV-027	IV catheter (22G / 24G)	11	80/20	0.977	-2.21	-0.81	0.9	57%
INV-028	Disposable gloves (box, 100 pcs)	82	80/20	0.964	-4.81	0.95	1.1	8%
INV-029	Gauze pads (sterile packs)	14	80/20	0.840	-2.36	0.73	0.4	21%

Code	Item	Days	Test	Train R2	Test R2 fitted	Test R2 anch.	MAE anch.	Rate err.
INV-030	Cotton balls / cotton rolls	6	in-sample	0.910	-	-	0.4	-
INV-031	70% alcohol (500 mL – 1L)	4	in-sample	0.921	-	-	0.3	-
INV-032	VetClean Anti-Tick & Flea Dog Sham	8	in-sample	0.823	-	-	0.8	-
INV-033	PetDerm Sensitive Skin Cat Shampoo	11	80/20	0.978	-5.05	-2.52	1.3	74%
INV-034	FungiCare Medicated Antifungal Sha	8	in-sample	0.964	-	-	0.3	-
INV-035	CoatSoft Pet Conditioner 200ml	7	in-sample	0.880	-	-	0.7	-
INV-036	FeshPaws Grooming Cologne (Baby Po	5	in-sample	0.960	-	-	0.3	-
INV-037	KnotFree Detangling Spray 150ml	7	in-sample	0.972	-	-	0.3	-
INV-038	Rabigen Anti-Rabies Vaccine 1ml (C	1	Insufficient data					
INV-039	Vanguard Plus 5-in-1 Vaccine for D	2	Insufficient data					
INV-040	Vanguard Plus 6-in-1 Vaccine for D	5	in-sample	0.922	-	-	0.4	-
INV-041	Felocell 4-in-1 Vaccine for Cats 1	5	in-sample	0.804	-	-	0.6	-
INV-042	Drontal	6	in-sample	0.961	-	-	0.3	-
ANA-1751D5	Rabies Vaccine	75	80/20	0.995	-0.81	0.55	3.1	33%
ANA-E5EED2	Distemper Combo Vaccine	52	80/20	0.990	-0.33	0.31	3.4	36%
ANA-ADD19A	Parvo Vaccine	47	80/20	0.983	-3.41	0.09	2.8	17%
ANA-D8D9BC	Bordetella Vaccine	61	80/20	0.991	-3.67	0.25	3.1	40%
ANA-944E58	Pet Shampoo	36	80/20	0.986	0.32	-0.63	2.9	43%
ANA-7AC8E2	Conditioning Spray	33	80/20	0.983	-2.03	0.78	1.1	14%
ANA-E85CA7	Ear Cleaner Solution	28	80/20	0.986	-4.43	0.55	1.0	23%
ANA-DE3CFA	Nail Clippers Set	18	80/20	0.973	-0.09	0.87	0.5	10%
ANA-14AB54	Premium Dog Food 3kg	69	80/20	0.990	0.04	0.35	5.0	38%
ANA-B11128	Cat Food Sachets	58	80/20	0.988	0.57	0.90	1.7	12%
ANA-C78F58	Vitamin E Drops	44	80/20	0.987	-0.58	-0.33	2.8	35%
ANA-C20F42	Probiotic Supplement	65	80/20	0.985	-4.69	0.71	2.2	24%
PRD-841633-8172	Heartgard Plus Chewables	1	Insufficient data					

In-sample: fewer than 10 days of usage, so the item is scored on the same days it learned from (no separate test). Slow movers with fewer than 3 days of usage stay "Insufficient data", which is the correct answer for them.

Chapter 5 - Summary, Conclusions, Recommendations

5.1 Summary

AutoVet's inventory forecast learns each item's usage per day from the clinic's usage history and estimates when stock will reach its minimum level. Because almost no usage was recorded from June 2026, 19 of 41 items could not be forecast. The missing weeks were filled with 1122 generated records whose rates, weekly pattern and growth were all learned from a full year of real records, with day-to-day variation drawn from a Poisson distribution. The forecast was then measured with R2, RMSE, MAE and an 80/20 time split, using the same formulas as the system's evaluation guide.

5.2 Conclusions

- Filling the gap made the forecast usable across the inventory: 37 of 41 items can now be forecast (from 22), and 24 can be tested on unseen days (from 5).
- The generated data behaves like real usage: the total rate equals the real rate, and its variation matches what a Poisson process predicts (dispersion index close to 1).
- The learned usage per day is a reasonable guide for planning stock: when started from the latest known total it beats the simple average (median MAE 2.78 vs 3.70 units), with a median rate error of 32%.
- The straight-line forecast, as currently placed, should not be judged by its training R2 alone: that number is always high for running totals and the "High" confidence label is therefore too optimistic.

5.3 Recommendations

- Place the forecast line at the latest known total (anchored), the way it is used for "days until stock-out", instead of using the fitted intercept.
- Base the confidence label on the test R2 or the rate error from the 80/20 split, not on the training R2.
- Record every real usage (sales and items used in services) consistently, so that the generated records can be removed as real history builds up. Command: php artisan inventory:seed-usage --purge (add --live for production).
- Re-run the benchmark every month (php artisan inventory:forecast-benchmark --real-only for real data only) and compare with this report.
- Consider a model that handles weekly and seasonal patterns (for example exponential smoothing) once at least a year of continuous real data exists.

5.4 Limitations and disclosure

- About 77% of the usage records (and about 40% of the units) in the last 365 days are generated. Results "after" therefore partly measure how well the model learns the clinic's own past pattern, not new real behaviour.
- Generated usage assumes the future looks like the reference year, with steady growth (capped at 3% a month). It cannot capture real events in the gap months (outbreaks, holidays, stock-outs).
- Items never used in the reference year received small, assumed rates based on their category.
- Generated records exist only in the usage history; stock levels, invoices and the transaction ledger are unchanged, so revenue and sales charts do not include them.

Suggested wording for the paper

> "Because the clinic recorded little inventory usage between June and October 2026, the gap was filled with simulated usage records. Their parameters (per-item daily rate, weekday weights and monthly growth) were estimated from one year of the clinic's real records (June 2025 to May 2026), and daily quantities were drawn from a Poisson distribution. Simulated records are labelled separately in the database and were not used to replace any real record."